

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

The Saving Account Management Module (SAMM) is the vital component of the proposed Online KADA System designed to provide members with a convenient, digital method of managing accounts. Members will be able to access a series of useful tools provided through SAMM to deposit funds more conveniently, transfer money, inquire about monthly installment details of loan amounts, view transaction history, and download receipts. These are intended to eliminate the inefficiencies of the manual system, with a user-friendly and accessible platform to provide a solution to meet the financial management needs of the members.

The necessity for SAMM arose from the inefficiencies and inconvenience of the existing manual system. KADA members now need to physically visit branches in order to make financial transactions. This process is not only time-consuming, but also inconvenient, particularly for members remote from branch offices. Additionally, the current system is not transparent because members are unable to view financial data in real time, for instance, transaction history or their repayment schedules. This transparency issue tends to create confusion and suspicion that erodes trust in the system.

Besides this, the absence of modern financial facilities in the manual system also provides more fuel to the problem. The members are not given digital receipts or have the tools to monitor their financial transactions, and therefore they cannot audit or organize their transactions effectively. SAMM wipes away these necessary drawbacks by providing a digital, open, and convenient solution where members are able to manage their finances more suitably and confidently.

Group Members' Name	Matric No	Module
Choh Jing Yi	A23CS0296	Financial Report Module
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Lim Yu Han	A23CS0241	Saving Account Management Module

1.2 Project Weakness or Improvement

Although the Saving Account Management Model (SAMM) is an improvement over other manual models, it does retain some serious flaws that detract from its overall utility. The most significant of these is the absence of automatic notifications. There are no alerts to users of critical financial events, like administrator approval or rejection of transfer requests. Consequently, members must log in and review their accounts individually in order to stay informed, which undermines the convenience and real-time efficiency an online platform should provide.

The other important limitation of SAMM is its inability to include personalization features for the users. The system does not have the function for users to sort or classify their transactions. Without the functionality to arrange financial activities, it is not easy for members to examine their accounts systematically. The inability to sort or classify transactions makes it difficult to analyze the behavior of finances, hence reducing the effectiveness of the system to end users who use it to track their financial performance.

Additionally, a major drawback is that SAMM lacks interactive reporting features. Users are not presented with visualizations of their financial data in the form of charts or graphs showing patterns of expenditure. Lack of visualized data provides the users with limited ability to comprehend their financial performance over different time spans. In turn, this shortcoming weakens their capacity to make well-informed choices on their saving and spending habits, hence reducing the module's overall effectiveness in relation to financial well-being.

1.3 Proposed Module

The Enhanced Saving Management Module (ESMM), an upgrade that integrates cutting-edge capabilities to offer a more thorough and user-focused financial management experience, has been suggested as a solution to SAMM's shortcomings. The addition of automatic notifications is one of the main enhancements to ESMM. Members will be notified in real time about significant financial events, such the approval or rejection of their transfer requests so they can now stay informed and involved at all times without having to manually check their accounts for updates.

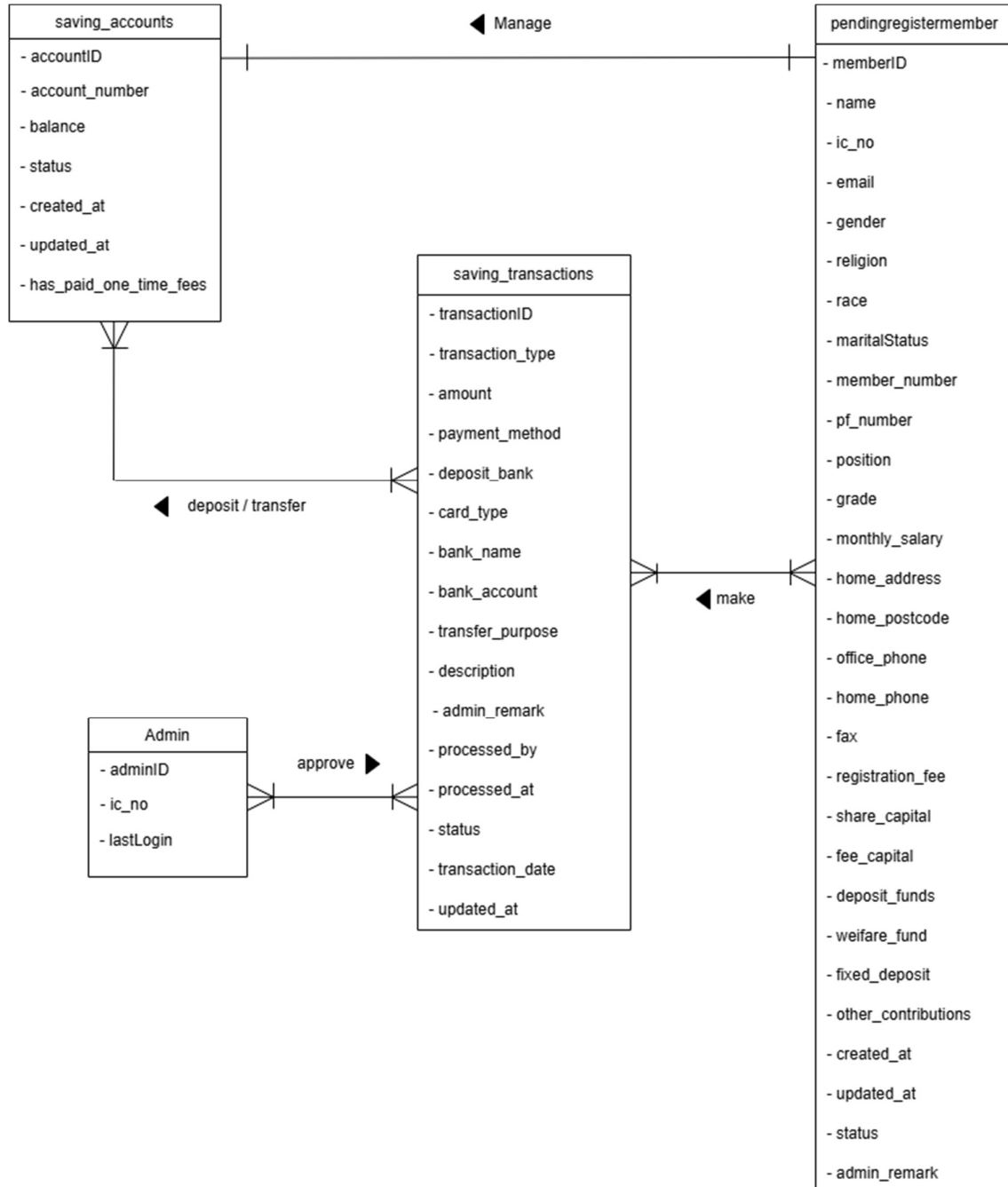
Another significant enhancement in ESMM is the introduction of transaction categorization and filtering capabilities. Members will now be able to classify their transactions into categories such as deposits, transfers or loan payments. This feature allows for better organization and easier recordkeeping, enabling users to review their financial activities more efficiently. Additionally, filters will be available to help members quickly access specific types of transactions, making the system more user-friendly and tailored to individual needs.

Dynamic financial reporting is also a centerpiece of ESMM's upgrades. The module will feature interactive tools such as charts and graphs that visualize transaction trends and spending patterns. These tools provide members with valuable insights into their financial behavior, helping them identify areas where they can improve their saving or spending habits. By offering a clear and comprehensive view of their financial performance, ESMM empowers members to make informed decisions that support their long-term financial goals.

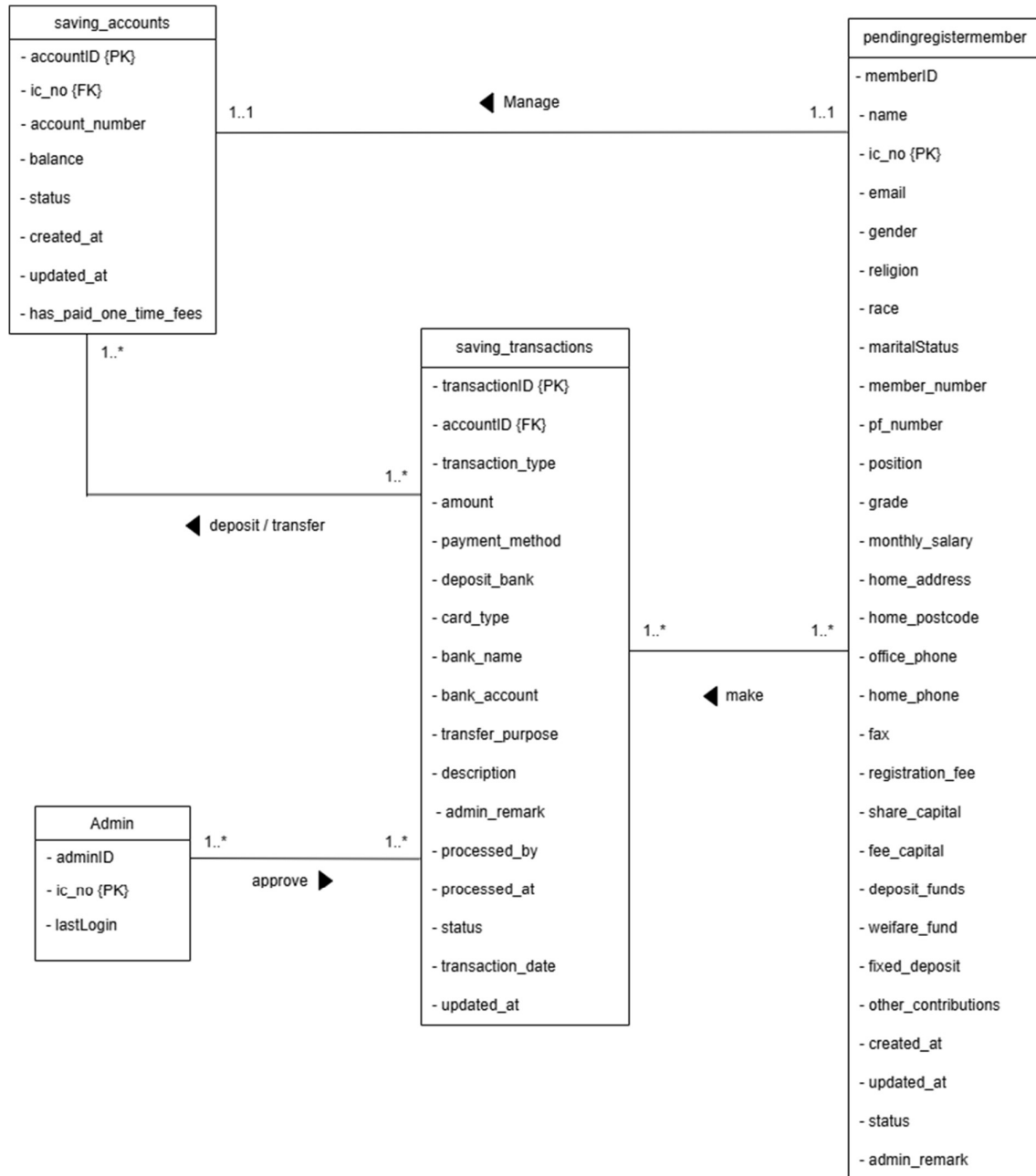
Ultimately, ESMM fosters a greater sense of control and confidence in financial management, ensuring that members have the tools they need to succeed.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary For Logical ERD

Entity Name	Attributes	Data Type & Length	Description	Null
pendingregist ermember	memberID	Int	Member's unique identifier	No
	name	Varchar (255)	Member's full name	No
	ic_no {PK}	Varchar (22)	Member's identification card number	No
	email	Varchar (30)	Member's email	No
	gender	Enum (male,female)	Member's gender	No
	religion	Varchar (50)	Member's religion	No
	race	Varchar (50)	Member's race	No
	marital_status	Enum (single, married, divorced, widowed)	Member's marital status	No
	member_number	Varchar (50)	Member's member number in kada	No
	pf_number	Varchar (50)	Member's personal file number in kada	No
	position	Varchar (100)	Member's working position in kada	No
	grade	Varchar (50)	Member's position grade in kada	No
	monthly_salary	Decimal (10,2)	Member;s monthly salary	No
	home_address	Text	Member's home address	No
	home_postcode	Varchar (10)	Member's home postcode	No
	home_state	Varchar (50)	Member's home state	No
	office_address	Text	Member's office address	No
	office_postcode	Varchar (10)	Member's office postcode	No
	office_phone	Varchar (20)	Member's office phone number	No
	home_phone	Varchar (20)	Member's phone number	No
	fax	Varchar (20)	Member's fax number	Yes
	registration_fee	Decimal (10,2)	Member's registration fee	No
	share_capital	Decimal (10,2)	Member's share capital	No
	fee_capital	Decimal (10,2)	Member's fee capital	No
	deposit_funds	Decimal (10,2)	Member's deposit funds	No
	welfare_fund	Decimal (10,2)	Member's welfare funds	No

	fixed_deposit	Decimal (10,2)	Member's fixed deposit	No
	other_contributions	Text	Member's other contribution	Yes
	created_at	Timestamp	Member's profile created time	Yes
	updated_at	Timestamp	Member's profile updated time	Yes
	status	Enum (pending, approve, rejected)	Member's application approval status	No
	admin_remark	Text	Admin remark on the approval status	Yes
admin	adminID	Int	Admin's unique identifier	No
	ic_no {PK}	Varchar (22)	Admin's identification card number	No
	lastLogin	Datetime	Admin's last login time	Yes
saving_accounts	accountID {PK}	Int	Unique saving account id	No
	ic_no {FK}	Varchar (22)	Member's identification card number	No
	account_number	Varchar (20)	Kada saving account number	No
	balance	Decimal (10,2)	Current balance of the saving account	No
	status	Enum (active, inactive)	Status of the saving account	No
	created_at	Timestamp	Account created time	No
	updated_at	Timestamp	Account updated time	No
	has_paid_one_time_fees	Boolean	Registration fee to become a member	No
saving_transactions	transactionID {PK}	Int	Unique transaction identifier	No
	account_id {FK}	Int	Member's saving account id	No
	transaction_type	Enum (deposit, transfer)	Type of the transactions	No
	amount	Decimal (10,2)	Transaction amount	No
	payment_method	Enum (fpx, card)	Payment method for the transaction	Yes
	deposit_bank	Varchar (50)	Bank name for transaction deposit	Yes

	card_type	Varchar (20)	Type of the card used for transaction deposit	Yes
	bank_name	Varchar (50)	Bank name for transaction transfer	Yes
	bank_account	Varchar (20)	Bank account for transaction transfer	Yes
	transfer_purpose	Varchar (50)	Purpose of the transaction transfer	Yes
	description	Text	Description for the transaction	No
	admin_remark	Text	Admin remark on approve transfer request	Yes
	processed_by	Int	Admin id that process the transfer request	Yes
	processed_at	Timestamp	Time of the transaction	Yes
	status	Enum (pending, approved, rejected)	Status of the transfer request	Yes
	transaction_date	Timestamp	Date of the transaction	Yes
	updated_at	Timestamp	Time where account last updated	Yes

2.4 Relational Database Schema

1. pendingregistermember (memberID, name, ic_no, email, gender, religion, race, maritalStatus, member_number, pf_number, position, grade, monthly_salary, home_address, home_postcode, office_phone, home_phone, fax, registration_fee, share_capital, fee_capital, deposit_funds, welfare_fund, fixed_deposit, other_contributions, created_at, updated_at, status, admin_remark)

Primary key: ic_no

Alternate key: email

2. admin (adminID, ic_no, lastLogin)

Primary key: ic_no

Alternate key: adminID

3. saving_accounts (accountID, ic_no, account_number, balance, status, created_at, updated_at, has_paid_one_time_fees)

Primary key: accountID

Foreign key: ic_no REFERENCES pendingregistermember (ic_no)

Alternate key: account_number

4. saving_transactions (transactionID, accountID, transaction_type, amount, payment_method, deposit_bank, card_type, bank_name, bank_account, transfer_purpose, description, admin_remark, processed_by, processed_at, status, transaction_date, updated_at)

Primary key: transactionID

Foreign key: accountID REFERENCES saving_accounts (accountID)

2.5 Normalization

1. pendingregistermember

First Normal Form (1NF):

- pendingregistermember (memberID, name, ic_no, email, gender, religion, race, maritalStatus, member_number, pf_number, position, grade, monthly_salary, home_address, home_postcode, office_phone, home_phone, fax, registration_fee, share_capital, fee_capital, deposit_funds, welfare_fund, fixed_deposit, other_contributions, created_at, updated_at, status, admin_remark)

primary key: ic_no

alternate key: email

Second Normal Form (2NF):

- pendingregistermember (memberID, name, ic_no, email)
- member_info (gender, religion, race, maritalStatus, member_number, pf_number, position, grade, monthly_salary, home_address, home_postcode, office_phone, home_phone, fax, registration_fee, share_capital, fee_capital, deposit_funds, welfare_fund, fixed_deposit, other_contributions, created_at, updated_at, status, admin_remark)

primary key: member_number

Third Normal Form (3NF):

- pendingregistermember (name, ic_no, email, gender, religion, race, maritalStatus, position, home_address, registration_fee, share_capital, fee_capital, deposit_funds, welfare_fund, fixed_deposit, other_contributions, created_at, updated_at, status, admin_remark)
- address_info (home_address, home_postcode, home_phone, fax)
- position_info (member_number, pf_number, position, grade, monthly_salary)

primary key: ic_no

primary key: home_address

primary key: position

2. admin

First Normal Form (1NF):

- admin (adminID, ic_no, lastLogin)

primary key: ic_no

alternate key: adminID

Second Normal Form (2NF):

- The primary key is (ic_no), all non primary attributes (adminID and lastLogin) are fully depend on (ic_no). The relation is already in 2NF.

Third Normal Form (3NF):

- lastLogin directly depends on primary key (ic_no), no transitive dependencies exist. The relation is already in 3NF.

3. Saving_accounts

First Normal Form (1NF):

- saving_accounts (accountID, ic_no, account_number, balance, status, created_at, updated_at, has_paid_one_time_fees)

primary key: accountID

foreign key: ic_no REFERENCES pendingregistermember (ic_no)

alternate key: account_number

Second Normal Form (2NF):

- The primary key is (accountID), there are no partial dependencies, the table is in 2NF.

Third Normal Form (3NF):

- All attributes depend directly on accountID, there are no transitive dependencies, the table is already in 3NF.

4. saving_transactions

First Normal Form (1NF):

- saving_transactions (transactionID, accountID, transaction_type, amount, payment_method, deposit_bank, card_type, bank_name, bank_account, transfer_purpose, description, admin_remark, processed_by, processed_at, status, transaction_date, updated_at)

primary key: transactionID

foreign key: accountID REFERENCES saving_accounts (accountID)

Second Normal Form (2NF):

- saving_profile (transactionID, accountID)

primary key: accountID

- transaction (transaction_type, amount, payment_method, deposit_bank, card_type, bank_name, bank_account, transfer_purpose, description, admin_remark, processed_by, processed_at, status, transaction_date, updated_at)
primary key: transaction_type

Third Normal Form (3NF):

- saving_transactions (transactionID, accountID, transaction_type, admin_remark, processed_by, processed_at, status, transaction_date, updated_at)
primary key: transactionID
- deposit_transaction (transaction_type, amount, payment_method, deposit_bank, card_type, description)
primary key: transaction_type
- transfer_transaction (transaction_type, amount, bank_name, bank_account, transfer_purpose, description)
primary key: transaction_type
- bank_details (payment_method, deposit_bank, card_type, bank_name, bank_account)
primary key: bank_account